

IPR UNIT-3

1. What are the patentable subject matters of an invention

A. 1. Patentable Subject Matter

- **Definition:** The invention must fall within categories recognized by patent law.
- **Includes:**
 - Process (method)
 - Machine
 - Article of manufacture
 - Composition of matter
 - Improvements thereof
- **Excludes** (India – Section 3 of Patents Act):
 - Laws of nature, abstract ideas, mathematical methods
 - Business methods, algorithms per se
 - Medical/surgical methods
 - Plants/animals (except microorganisms)
 - Traditional knowledge

2. Novelty

- **Definition:** The invention must be **new**—not disclosed anywhere in the world before the filing date.
- **Test:** Compare invention against **prior art** (existing knowledge, publications, patents).
- **Key Point:** Even a single prior disclosure destroys novelty.
- **Example:** A new drug molecule not previously published or patented.

3. Inventive Step / Non-Obviousness

- **Definition:** The invention must involve a **technical advance** or **economic significance** that is not obvious to a person skilled in the art.
- **Test:** Would a skilled professional, using common knowledge, find the invention obvious?
- **Example:**
 - **Obvious:** Changing color of a known product.
 - **Non-obvious:** A new drug delivery system that improves bioavailability significantly.

4. Capable of Industrial Application

- **Definition:** The invention must be usable in industry—manufacturable or applicable in practice.
- **Requirement:** It should produce a **repeatable, useful result**.
- **Example:**
 - Patentable: A new type of solar panel that generates electricity.
 - Not patentable: A purely theoretical formula without practical application.

2. Explain about the Intellectual Property Appellate Board (IPAB)

A. Intellectual Property Appellate Board (IPAB)

1. Establishment

- Constituted by the Central Government on **15 September 2003**.
- Functioned under the Ministry of Commerce and Industry.
- Created to reduce burden on civil courts and provide specialized expertise in IP disputes.

2. Jurisdiction & Functions

- Heard appeals against decisions/orders of:
 - Registrar of **Trade Marks** (under Trade Marks Act, 1999).
 - Registrar of **Geographical Indications** (GI Act, 1999).
 - Controller of **Patents** (Patents Act, 1970).
 - Copyright Board (later merged into IPAB).
- Could also handle rectification/cancellation of IP registrations.

3. Composition

- **Chairperson** (usually a retired judge).
- Judicial Members and Technical Members (experts in patents, trademarks, copyright).
- Ensured both legal and technical expertise in decision-making.

4. Abolition (2021)

- The **Tribunals Reforms Ordinance, 2021** abolished IPAB along with several other tribunals.
- Reason: Streamlining tribunals, reducing delays, and strengthening High Courts' role.
- Consequence: All pending IPAB cases transferred to respective High Courts.

3. Explain Ownership and Transfer of Patent

A. Patent ownership, which grants exclusive rights to an invention, can be transferred through assignment (full ownership transfer), licensing (permission to use the invention under specified terms), or transmission (transfer due to inheritance or operation of law like a merger). Assignments involve a formal written agreement, while licenses can be exclusive or non-exclusive, and transmissions occur through legal processes like wills or company reorganizations.

Patent Ownership

1. Who owns it

2. What are the rights

Ownership is Transferred

There are three main ways to transfer patent rights:

1. Assignment:

- This is a complete transfer of all patent rights from the original owner (assignor) to a new owner (assignee).

2. Licensing:

- A license grants another party (licensee) permission to make, use, or

The Patent Cooperation Treaty (PCT) is an international treaty established in 1970. It enables inventors to seek patent protection in multiple countries through a single international application, rather than filing separate applications in each country.

Purpose and Benefits

- Simplifies the process of filing patents across borders.
- Saves time, effort, and cost by avoiding the need to prepare multiple applications in different languages and formats.
- Offers additional time to applicants to decide in which countries they want to pursue patent protection.

sell the patented invention without transferring ownership

3. Transmission:

- This occurs when ownership is transferred due to an event other than a voluntary agreement

4. Illustrate Compulsory Licensing of patent along with an example

A. Compulsory licensing is a provision under patent law that allows a government to authorize a third party to produce a patented product or use a patented process without the consent of the patent owner. It is typically invoked to protect public interest, especially in cases of public health emergencies or when patented products are not available at affordable prices.

Legal Basis (India) Under the Indian Patents Act, 1970:

- Section 84 allows any person to apply for a compulsory license after three years from the date of patent grant.
- Grounds include:
 - Reasonable requirements of the public not being met
 - Non-affordability of the patented invention
 - Non-working of the patent in India
- Section 92 allows the government to issue compulsory licenses in cases of national emergency or public health crises.

Example: Bayer vs. Natco (India, 2012)

- Patent holder: Bayer AG (Germany)
- Drug: Nexavar (sorafenib tosylate), used for treating liver and kidney cancer
- Issue: Bayer priced the drug at ₹2.8 lakhs/month, unaffordable for most Indian patients
- Action: Natco Pharma applied for a compulsory license
- Outcome:
 - License granted under Section 84
 - Natco sold the drug at ₹8,800/month
 - Natco paid 6% royalty to Bayer
 - First-ever compulsory license granted in India

5. Discuss about the Patent Cooperation Treaty (PCT)

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single international application, rather than filing separate applications in each country.

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- Offers additional time to applicants to decide in which countries they want to pursue patent protection.
- Used extensively by major corporations, universities, and research institutions.

PCT Filing Process (Step-by-Step)

1. File local application in your home country.
2. Submit PCT application.
3. Check for defects in the application.
4. Receive International Search Report.
5. International publication of the application.
6. Conduct International search.
7. Obtain International Preliminary Report.
8. Enter national phase in designated countries for final patent grant.

Patent Term Under PCT

For applications filed under the national phase of the PCT, the patent term is 20 years from the international filing date

6. Examine the requirements that determine what can be patented in India

A. To be patentable in India, an invention must satisfy three core criteria: novelty, inventive step, and industrial applicability. Additionally, it must not fall under the exclusions listed in Sections 3 and 4 of the Patents Act, 1970.

Core Patentability Criteria (Section 2 of the Patents Act, 1970)

1. Novelty

- The invention must be *new*—not disclosed in any prior publication, use, or knowledge anywhere in the world before the filing date.

2. Inventive Step (Non-obviousness)

- The invention must involve a *technical advancement* or *economic significance* or both, and must not be obvious to a person skilled in the art.

3. Industrial Applicability

- The invention must be *capable of being made or used* in an industry. It should have practical utility.

List of Non-Patentable Items

1. Creations that harm public order, morals, health, or the environment, or are deemed trivial.
2. Scientific discoveries that do not qualify as inventions.
3. Mere variations or alternatives of known substances.
4. Agricultural or horticultural methods and processes.

5. Methods for medical diagnosis, treatment, or surgery for humans or animals.
6. Living organisms, except microorganisms.

7. Examine the exclusive rights of a patent holder and its limitations

A. Exclusive Rights of a Patent Holder

A patent grants the holder a set of exclusionary rights under national law. These include:

1. Right to exclude others from making the patented invention.
2. Right to exclude others from using the invention for commercial or private purposes.
3. Right to exclude others from selling or offering for sale the invention.
4. Right to exclude others from importing the invention into the jurisdiction of the patent.
5. Right to license or assign the patent to third parties for consideration.

These rights are negative in nature—they do not confer a positive right to use the invention, but rather to prevent others from unauthorized use. The standard term is 20 years from the filing date (for utility patents), subject to maintenance and renewal.

Limitations on Patent Rights

Patent exclusivity is subject to several statutory, equitable, and public interest limitations:

1. Territorial Limitation Patent protection is confined to the country of grant. Enforcement requires separate filings in each jurisdiction.
2. Time Limitation Rights expire after the statutory term (usually 20 years), after which the invention enters the public domain.
3. Subject Matter Exclusion Certain categories—abstract ideas, natural laws, algorithms per se, and inventions contrary to morality—are excluded from patentability.
4. Government Use and Compulsory Licensing Governments may authorize use of patented inventions without consent for public purposes (e.g., health emergencies), subject to reasonable compensation.
5. Research and Experimental Use Exception Non-commercial academic or scientific use of a patented invention is generally exempt from infringement.
6. Bolar Exception Allows generic manufacturers to use patented drugs for regulatory approval before expiry, enabling market entry immediately after patent expiration.
7. Exhaustion of Rights (First Sale Doctrine) Once a patented product is sold lawfully, the patentee cannot control its resale or downstream use.
8. Parallel Imports Some jurisdictions permit import of patented goods sold abroad with the patentee's consent, limiting control over international distribution.
9. Use on Foreign Vessels or Aircraft Temporary use of patented inventions on foreign vehicles passing through a country may be exempt from infringement.

8. Discuss in detail the timeline of Patent laws in India and some of the provisions of the Patent Act, 1970

A. 1911 – Indian Patents and Designs Act

- First consolidated statute governing patents and designs.
- Allowed both product and process patents.
- Administered by the Controller of Patents.
- Remained in force until 1972.

1949 – Tek Chand Committee Report

- Recommended reforms to align patent law with public interest.
- Highlighted misuse of product patents in pharmaceuticals.

1957 – Justice Rajagopala Ayyangar Committee Report

- Proposed a new patent law focused on process patents.
- Emphasized indigenous innovation and access to medicines.
- Basis for the Patents Act, 1970.

1970 – Patents Act, 1970

- Replaced the 1911 Act.
- Came into force on 20 April 1972.
- Allowed only process patents for food, medicine, and chemicals.
- Strengthened compulsory licensing and public interest safeguards.

1999 – First Amendment (TRIPS compliance)

- Introduced Exclusive Marketing Rights (EMRs).
- Established a “mailbox” system for pharmaceutical and agrochemical product patent applications.

2002 – Second Amendment

- Redefined “invention” and “inventive step.”
- Strengthened opposition procedures and compulsory licensing.
- Allowed product patents in all fields except pharma and agrochemicals.

2005 – Third Amendment

- Enabled product patents for pharmaceuticals and agrochemicals.
- Fully TRIPS-compliant.
- Introduced pre-grant and post-grant opposition.
- Removed EMRs and mailbox system.

II. Key Provisions of the Patents Act, 1970 (as amended)

Section 2 – Definitions

- Defines “invention,” “patentee,” “inventive step,” “new invention,” etc.

Section 3 – Non-patentable inventions

- Excludes:
 - Frivolous inventions
 - Inventions contrary to public order or morality
 - Mere discoveries of natural substances
 - Business methods, algorithms, and mathematical methods
 - Traditional knowledge

Section 6 – Who can apply

- Inventor, assignee, or legal representative.

Section 10 – Specification

- Requires full disclosure of the invention and best method of performance.

Section 11A – Publication of application

- Patent applications are published after 18 months unless early publication is requested.

9. Summarize product patent and process patent and WTO's stand on product and process patents

A. Product Patent

- Grants protection to the **actual product**, regardless of how it is made.
- Infringement occurs if anyone makes, uses, sells, or imports the same product without authorization.
- Example: A patented drug molecule—no one can manufacture or sell it, even using a different method.

Process Patent

- Grants protection to the **method or process** of making a product.
- Others may produce the same product using a **different process** without infringing the patent.
- Example: A patented method to synthesize a chemical—others can use alternative methods to make the same chemical.

WTO's Stand (TRIPS Agreement, 1995)

- The **TRIPS Agreement** (Trade-Related Aspects of Intellectual Property Rights) mandates that:
 - All WTO members must provide **both product and process patents** in all fields of technology.
 - Minimum patent term must be **20 years** from the filing date.
 - Patent rights must be **non-discriminatory** with respect to technology, product origin, or field of invention.
- **India's compliance:**
 - Initially allowed only **process patents** for food, drugs, and chemicals (under the Patents Act, 1970).
 - Amended its law in **2005** to allow **product patents** in all sectors, including pharmaceuticals and agrochemicals, to comply with TRIPS.

10. Express Software protection and Computer related Innovations in detail

A. Software protection in India is primarily governed by copyright law, while computer-related inventions (CRIs) may be patentable if they meet specific technical criteria. The Patents Act, 1970 excludes “per se” software, but allows embedded software innovations under certain conditions.

1. Software Protection in India

Copyright Protection (Primary Mode):

- Governed by the **Copyright Act, 1957**.
- Software is protected as a **literary work**.
- Protection covers **source code, object code, structure, and user interface** (if

original).

- **Automatic protection** upon creation; registration is optional but recommended for enforcement.
- Duration: **60 years** from the year following the author's death (for published works).

Limitations:

- Copyright does **not protect underlying ideas, algorithms, or functionality**.
- It only prevents copying of expression (code), not independent development.

2. Patent Protection for Computer-Related Inventions (CRIs)

Legal Framework:

- Governed by the **Patents Act, 1970** and **Patent Rules, 2003**.
- **Section 3(k)** excludes "a mathematical or business method or a computer program per se or algorithms" from patentability.

Interpretation of "per se":

- The term "per se" implies that **pure software** (standalone programs) is not patentable.
- However, **software with technical application or hardware integration** may be patentable.

Patentable CRIs (if they meet 3-part test):

1. **Novelty** – Not disclosed in prior art.
2. **Inventive Step** – Not obvious to a skilled person.
3. **Industrial Applicability** – Capable of being made or used in industry.

Examples of Acceptable CRIs:

- Software controlling hardware (e.g., washing machine logic).
- Embedded systems (e.g., anti-lock braking software).
- Technical solutions using software (e.g., image compression algorithms with hardware implementation).

Non-Patentable Examples:

- Pure business methods implemented via software.
- Standalone mobile apps without technical effect.
- Algorithms without hardware linkage.

3. Judicial and Policy Developments

- **Controller's Guidelines (2017, revised 2019):** Clarify that CRIs must demonstrate a **technical contribution** beyond mere automation.
- **Case Law:**
 - *Ferid Allani v. Union of India (2019)* – Delhi High Court held that software with technical effect may be patentable.
 - *Ericsson v. Intex* – Asserted patent rights over telecom software.

11. Summarize step-by-step procedure to file a patent in India

A. Step 1: Patent Search (Optional but Recommended)

- Conduct a **prior art search** to check if the invention is novel and patentable.
- Helps avoid rejection due to lack of novelty or obviousness.

Step 2: Drafting the Patent Application

- Prepare either:
 - **Provisional Specification** (if invention is incomplete)
 - **Complete Specification** (if invention is fully developed)
- Must include: title, field of invention, background, summary, drawings (if any), claims, and abstract.

Step 3: Filing the Application

- File with the **Indian Patent Office (IPO)** via Form 1.
- Include Form 2 (specification), Form 3 (statement of foreign filings), and Form 5 (declaration of inventorship).
- Can be filed online or physically at one of the four IPOs (Delhi, Mumbai, Chennai, Kolkata).

Step 4: Publication of Application

- Application is published after **18 months** from the filing date.
- Early publication can be requested via Form 9.

Step 5: Request for Examination

- Must be filed within **48 months** of the priority date using Form 18.
- The application enters the examination queue.

Step 6: Examination and Office Actions

- Examiner reviews patentability (novelty, inventive step, industrial applicability).
- Issues **First Examination Report (FER)** with objections, if any.
- Applicant must respond within **6 months**, extendable by 3 months.

Step 7: Pre-grant Opposition (if any)

- Any person may oppose the grant after publication but before grant (Section 25(1)).

Step 8: Grant of Patent

- If objections are resolved and no opposition is sustained, patent is granted.
- Patent certificate is issued and details are published in the Patent Journal.

Step 9: Post-grant Opposition (if any)

- Within **12 months** from grant, any person may file opposition (Section 25(2)).

Step 10: Renewal and Maintenance

- Annual renewal fees must be paid from the **third year onward** to keep the patent in force.
- Maximum term: **20 years** from filing date.

12. Discuss revocation of patent along with a relevant case

A. Revocation of patents Revocation, in layman terms, means the act of taking back something that has already been granted. The same happens to the patent, which even though once granted can't be treated as permanent. People, other than the patent holder, are provided with a right of filing a revocation petition if they have issues due to someone's patent right.

The following provide ground for revocation of the patent:

- If an identical invention has already been protected by the patent right with an earlier filing date.

- If the person provided with the patent is not the original owner and has by misrepresentation or fraud obtained the patent protection, the original owner can seek the remedy of patent revocation.
- An obvious invention can't be shed by patent protection.
- A patent can be revoked if granted to a person who is not entitled under the patent act 1970.
- If the invention fails to provide the service as specified in specification while applying for the patent.
- The inventor is bound to disclose all the information required by the section and if such inventor fails or misrepresent any of the asked information, the patent shall be revoked.
- If the invention does not meet the standard of Section 2(1) (j) of the Patent Act 1970, the patent granted on such invention can be revoked.
- If an invention lacks newness about prior public use or prior knowledge in such a scenario the patent can be revoked.
- If any inventor fails to comply with provided compliances his patent may be revoked.
- The inventions that come under the umbrella of Section 3 of the Patent Act, 1970, are not eligible to be patented.
- If the invention claimed by the inventor is already in the knowledge of any local or native community in India.